

ECA15 TRANSMISSION LINE & WAVEGUIDES

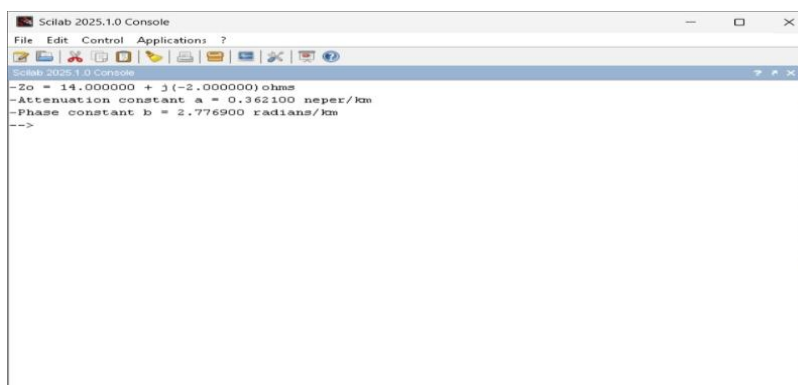
EXP 1

Estimation of Characteristic Impedance (Z_0), and Propagation Constant (γ) of a Transmission Line from Primary Constants

Test Case 1:

```
clear; clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*%pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

Output :



```
Scilab 2025.1.0 Console
File Edit Control Applications ?
- Zo = 14.000000 + j(-2.000000) ohms
- Attenuation constant a = 0.362100 neper/km
- Phase constant b = 2.776900 radians/km
-->
```

Test Case 2:

```
clc;
clear;
close;

// Input Values
R = 10;      // Resistance in Ohms
L = 3.5e-3;  // Inductance in Henry
G = 4e-6;    // Conductance in Siemens
C = 16e-6;   // Capacitance in Farad
f = 2.5e3;   // Frequency in Hz

// Angular Frequency
w = 2 * %pi * f;

// Series Impedance
Z = R + (%i * w * L);

// Shunt Admittance
Y = G + (%i * w * C);

// Characteristic Impedance
Zo = sqrt(Z / Y);

disp("Characteristic Impedance (Zo)");
disp(Zo);

// Propagation Constant
gamma = sqrt(Z * Y);

// Attenuation Constant
alpha = real(gamma);
disp("Attenuation Constant (Alpha)");
```

```
disp(alpha);
```

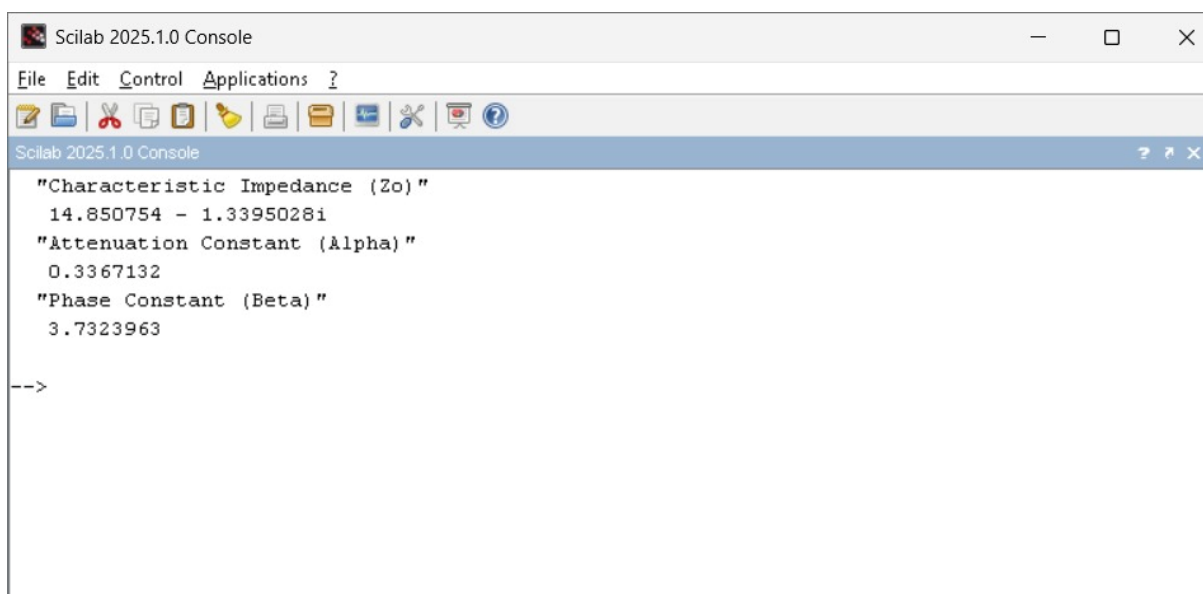
```
// Phase Constant
```

```
beta = imag(gamma);
```

```
disp("Phase Constant (Beta)");
```

```
disp(beta);
```

Output:



The image shows a screenshot of the Scilab 2025.1.0 Console window. The window has a title bar with the Scilab logo and the text "Scilab 2025.1.0 Console". Below the title bar is a menu bar with "File", "Edit", "Control", "Applications", and "?". Below the menu bar is a toolbar with various icons. The main area of the window displays the output of the code, which is as follows:

```
"Characteristic Impedance (Zo)"  
14.850754 - 1.3395028i  
"Attenuation Constant (Alpha)"  
0.3367132  
"Phase Constant (Beta)"  
3.7323963  
-->
```